Veering and Strong Coupling Effects in Structural Dynamics

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Mode veering is the phenomenon associated with the eigenvalue loci for a system with a variable parameter: two branches approach each other and then rapidly veer away and diverge instead of crossing. The veering is accompanied by rapid variations in the eigenvectors. In this paper, veering in structural dynamics is analyzed in general terms. First, a discrete conservative model with stiffness, mass, and/or gyroscopic coupling is considered. Rapid veering requires weak coupling: if there is instead strong coupling then there is a slow evolution of the eigenvalue loci rather than rapid veering. The uncoupled-blocked system is defined to be that where all degrees-of-freedom (DOFs) but one are blocked. The skeleton of the system is the loci of the eigenvalues of the uncoupled-blocked system as the variable parameter changes. These loci intersect at certain critical points in the parameter space. Following a perturbation analysis, veering is seen to comprise rapid changes of the eigenvalues in small regions of the parameter space around the critical points: for coupling terms of order ε veering occurs in a region of order ε around the critical points, with the rate of change of eigenvalues being of order ε^{-1}. This is accompanied by rapid rotations in the eigenvectors. The choice of coordinates in the model and application to continuous systems is discussed. For nonconservative systems, it is seen that veering also occurs under certain circumstances. Examples of 2DOFs, multi-DOFs (MDOFs), and continuous systems are presented to illustrate the results.
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1 Introduction

Mode veering typically refers to the phenomenon associated with the eigenvalue loci for a system with a variable parameter: two branches approach each other and then rapidly veer away and diverge instead of crossing. The veering is accompanied by rapid variations in the eigenvectors. Veering has been observed in many systems, primarily in the detailed analysis of specific systems. The aim of this study is to present a general analysis of the conditions under which veering occurs, and why eigenvalue loci veer rather than cross.

Some of the first evidence of coupling between modes was provided by Webster [1], Petyt and Fleisxer [2], and Nair and Durvasula [3]. In Ref. [3], frequency loci for plates with varying geometric parameters, geometry, stiffness, and boundary conditions were discussed. It was shown that for symmetric configurations the frequency curves diverge from each other. The authors referred to this behavior as "quasidegeneracies." Later Leissa [4], referring to the eigenfunctions, described the phenomenon using these words: "…must undergo violent change—figuratively speaking, a dragonfly one instant, a butterfly the next, and something indescribable in between. Thus the results appear strange from an aesthetic point of view." He then stated that "In numerous places in the literature of eigenvalue problems of mathematical physics one finds curves which approach each other and suddenly veer away. The author postulates that this ugly behavior may be the result of approximation in the representation of physical reality." Leissa [4], described the phenomenon of the eigenfunctions as an 'ulgy' behavior. Since then, the phenomena has been commonly referred to as veering since, under certain coupling conditions, two or more eigenvalue loci of a system with a varying parameter veer away and diverge instead of crossing—a schematic representation, described in more detail in Sec. 2, is given in Fig. 1: the abscissa and ordinate represent the variable parameter and system eigenvalues, respectively. The solid lines show the eigenvalues, which veer rather than cross, while the inlays show the eigenvectors, which rotate rapidly around the point where veering occurs.

Although Leissa showed that, for continuous models, curve veering can be an "aberration" introduced when natural frequencies and mode shapes are obtained using approximate methods associated with discretized models, veering has been verified later using exact solutions, and it is now commonly accepted that curve veering in continuous models is indeed a "representation of physical reality." One of the earlier veering criteria was formulated by Perkins and Mote [5]. In Ref. [5], a perturbation technique was applied in a straightforward way to analyze the dependence of the eigenvalue loci on a system parameter, both in the case of continuous and discrete systems. In particular, in Ref. [5], it was demonstrated that veering could not be an aberration. Later, Pierre [6], used a modified perturbation approach to overcome failings of Perkins and Mote's technique in the vicinity of the point of maximum coupling. A number of modified perturbation approaches

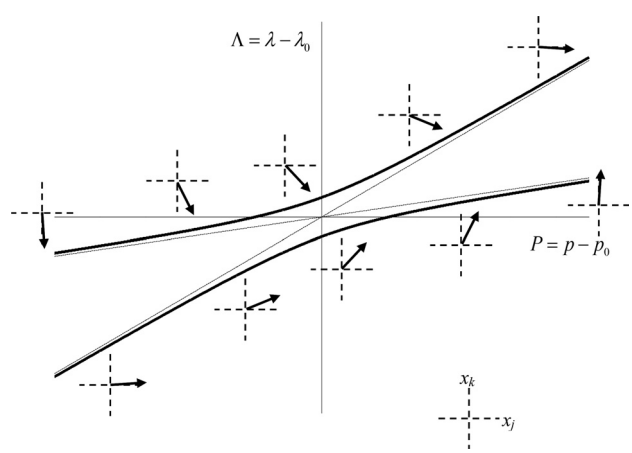

Fig. 1 Veering: eigenvalues of - - - - - uncoupled-blocked and ——— coupled DOFs x_j and x_k. Inlays show directions of eigenvectors in (x_j, x_k) plane.

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have been proposed since then, and veering phenomena in dynamics have been treated in many papers.

Veering also involves rapid changes in the eigenvectors, which rapidly rotate and do not cross. This aspect has been theoretically studied for example in Ref. [7], where it was shown how the eigenvalues can vary as a parameter of a Hamiltonian system changes. In dynamics mode veering was studied in Ref. [8] for a discrete 3DOF system. An analysis of veering using a geometric approach was given by Triantafyllou and Triantafyllou [9]. A geometric formulation of the problem was also given by Arnold [10] in terms of ellipsoids associated to positive definite symmetric eigenvalue problems, and later by Vidoli and Vestroni [11]. Evidence of the veering phenomena was also seen, for example, in pressure vessels [12] and cable dynamics [13] and recently experimental demonstrations of veering were given in Refs. [14,15]. Other examples of recent studies on veering include the vibrations of a spinning beam with a rigid body attached at its free-end [16], and a centrifugal pendulum vibration absorber [17]. Veering in bladed-disk assemblies was studied in Ref. [18], where it was seen that the effects of blade mistuning reduced the veering effects associated with mode localization; a perturbation method for the prediction of the forced response of mistuned bladed disks was recently illustrated in Ref. [19].

Coupling between modes and veering occurs in structural waveguides such as rods, plates, and cylinders. This is particularly true at higher frequencies, when the wavelengths of interest become small compared to the length of the structure. As examples, coupling between bending and torsional motion of uniform beams was studied by Timoshenko et al. [20], and later by Bishop and Pierce [21]. Veering was also seen to occur in the nonlinear response of a curved beam with imperfections [22]. Continuous structures with added subsystems and systems with centrifugal forces can exhibit mode veering. Examples abound in the literature, in particular concerning simple structural components such as beams and plates carrying spring-mass systems, e.g., see Ref. [23]. The phenomenon has also been observed in the vibrations of disordered structures. The presence of small irregularities in weakly coupled periodic systems affects the system behavior in terms of vibration mode and energy localization (vibration modes are "confined" to a limited region of the system) associated with veering of the frequency loci. In the context of structural vibration, one early study where mode localization was discussed, was given by Hodges [24]. The relation between veering and mode localization in periodic structures perturbed by a disordered parameter was investigated by Pierre in Ref. [25], although some of the Pierre's results were later criticized by Chen and Ginsberg [26]. Veering of frequency loci associated with mode localization was also studied and quantified by Lee and Mote [27] for a translating string coupled to a spring-mass-damper constraint, where frequency loci and mode localization were evaluated as functions of the constraint location. Recently, a quantification of vibration localization in periodic structures has been proposed in Ref. [28]. It has also been shown that mode localization occurs in weakly coupled nonlinear structures, e.g., see Ref. [29]. In the last decade, there have been interesting applications of energy localization associated with eigenvalue veering for sensing purpose, in particular concerning ultrasensitive resonant microelectromechanical systems (MEMS) transducers, e.g., see Refs. [30,31].

A general formulation of the veering phenomena is of practical interest in many applications such as modal analysis and modal parametrization, model updating, stability analysis, optimization, MEMS transducers, etc. However, most of the published work on veering concerns the analysis of specific systems and involves analytical or numerical studies on a case-by-case basis. Apart from a few classical works on discrete systems cited earlier, e.g., see Refs. [5–11], few theoretical studies have attempted to address the problem in a general way; one example is Ref. [32] where meta-modeling techniques were applied for the prediction of mode crossing and mode veering. This provides the motivation for the present work.

The purpose of this paper is to analyze veering in structural dynamics in general terms and describe the qualitative features of a class of coupled structures and the circumstances under which veering occurs. Starting from a discrete model, a definition of weak coupling is given—rapid veering requires weak coupling. The uncoupled-blocked system is defined. The skeleton of the system is the loci of the eigenvalues of the uncoupled-blocked system as the variable parameter changes. These loci intersect at certain critical points in the parameter space. Following a perturbation analysis, veering is seen to occur for weak coupling in the neighborhood of these critical points.

Section 2 concerns discrete, conservative systems. First, weak stiffness coupling is considered, followed by the case of general conservative (mass, stiffness, and gyroscopic) coupling. The choice of coordinates in the model and the application to continuous systems is considered in Sec. 3. Section 4 concerns nonconservative systems, and it is seen that veering also occurs in damped systems under certain circumstances. Section 5 presents numerical examples—2DOF systems, multi-DOFs, and continuous systems—while Sec. 6 concludes the paper.

2 Veering in Discrete, Conservative Systems

The equations of motion of free vibration for an undamped MDOF system can be written in discrete form as

$$[\mathbf{K}(\mathbf{p}) - \lambda\mathbf{M}(\mathbf{p})]\mathbf{x} = 0 \tag{1}$$

where $\mathbf{K}$ and $\mathbf{M}$ are the mass and stiffness matrices, $\mathbf{p}$ is a vector of variable or random parameters, the eigenvalues λ are the squares of the natural frequencies, and $\mathbf{x}$ is a vector of DOFs. These may be physical DOFs, the modal coordinates of the system calculated for some specific value of $\mathbf{p}$, the amplitudes in an assumed mode model, or perhaps a truncated set of any such coordinates. The eigenvalues are the roots of $D = \det(\mathbf{K}(\mathbf{p}) - \lambda\mathbf{M}(\mathbf{p})) = 0$. The eigenvectors $\boldsymbol{\phi}$ of Eq. (1) are the mode shapes. As the parameters vary, the eigenvalues follow loci in the $(\mathbf{p}, \lambda)$ space. Veering is observed to occur when two eigenvalues approach closely as a parameter increases and then veer apart rapidly, with consequent rapid changes in the eigenvectors. The eigenvectors associated with each branch of the spectrum are in essence exchanged.

2.1 Weak Coupling, the Uncoupled System and the Skeleton of the System. Veering is the term used to denote *rapid* changes in the eigenproperties. As will be seen below, this is associated with weak coupling.

The mass and stiffness matrices can be written as

$$\mathbf{M} = \mathrm{diag}(M_{jj}(\mathbf{p})) + \mathbf{M}_1(\mathbf{p}); \quad \mathbf{K} = \mathrm{diag}(K_{jj}(\mathbf{p})) + \mathbf{K}_1(\mathbf{p}) \tag{2}$$

where diag(•) represents a diagonal matrix, and the diagonal elements of $\mathbf{M}_1$ and $\mathbf{K}_1$ are zero. *Weak coupling* is defined to be the case, where the off-diagonal terms are small compared to the diagonal terms such that

$$\frac{M_{1,ij}}{\sqrt{M_{ii}M_{jj}}} = O(\varepsilon); \quad \frac{K_{1,ij}}{\sqrt{K_{ii}K_{jj}}} = O(\varepsilon) \tag{3}$$

where $\varepsilon \ll 1$.

The *uncoupled* system is defined here by *blocking* all DOFs apart from each one in turn, i.e., it is the system formed by setting the off-diagonal terms to zero. The uncoupled-blocked system matrices are diagonal with the jth term being $d_{jj} = K_{jj} - \lambda M_{jj}$. The *uncoupled* eigenvalues are therefore $\lambda_j^0 = K_{jj}(\mathbf{p})/M_{jj}(\mathbf{p})$ while the eigenvectors are $\boldsymbol{\phi}_j^0$, all of whose elements are zero except one.

The loci of the uncoupled-blocked eigenvalues form the *skeleton* of the system. Using the blocked definition of the uncoupled system, it is seen below that veering occurs around the *critical*

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points in the skeleton, namely those points where the branches of the skeleton intersect. This is illustrated in Fig. 1, where the dashed lines show the skeleton and the critical point is at the origin of the coordinates (P, Λ), defined below.

2.2 Perturbation Analysis in the Neighborhood of a Critical Point. Consider an n DOF, weakly coupled system. Expanding the determinant D in increasing powers of ε gives, in terms of decreasing magnitude

$$O(D) = O(d^n) + O(d^{n-2}\varepsilon^2) + O(d^{n-3}\varepsilon^3) + \cdots O(\varepsilon^n) \tag{4}$$

with d representing the diagonal terms (further details can be found in the Appendix). Far from any critical point all but one of d_{jj} are $O(1)$, and the roots of D are

$$\lambda_j = \lambda_j^{(0)} + O(\varepsilon^2) \tag{5}$$

and are hence perturbations of order ε^2 from the uncoupled-blocked values.

Now, consider the behavior around the critical point where $\lambda_j^{(0)}(p) = \lambda_k^{(0)}(p)$, i.e., when d_{jj} and d_{kk} are both equal to zero (the case of three intersecting eigenvalues is not considered). Close to this point the diagonal terms d_{jj} and d_{kk} are $\leq O(\varepsilon)$. The determinant D is then approximately

$$D = (d_{jj}d_{kk} - d_{jk}d_{kj})\left(\prod_{i=1,n;\neq j,k} d_{ii}\right) \tag{6}$$

where the terms in the product are $O(1)$. Veering occurs between branches j and k. A full analysis becomes lengthy and complicated due to the number of possible parameters and coupling conditions. For simplicity a stiffness-coupled system is considered first, followed by the case of general conservative coupling.

2.2.1 Stiffness-Coupled System. As a specific example, consider a weakly stiffness-coupled MDOF system with a single, variable stiffness parameter p. The eigenvalues are found from the solutions to

$$|\mathbf{K}(p) - \lambda\mathbf{M}| = 0 \tag{7}$$

with $\mathbf{M}$ diagonal and the off-diagonal elements of $\mathbf{K}$ being $\leq O(\varepsilon)$.[1] The uncoupled-blocked eigenvalues are $\lambda_j^0 = K_{jj}(p)/M_{jj}$. Far from any critical point the characteristic equation is

$$\prod_{jj=1}^{n}(K_{jj}(p) - \lambda M_{jj}) + O(\varepsilon^2) = \prod_{jj=1}^{n}\left(\lambda_j^0(p) - \lambda\right)M_{jj} + O(\varepsilon^2) = 0 \tag{8}$$

and the eigenvalues λ are perturbations of $O(\varepsilon^2)$ from the uncoupled-blocked values λ^0.

2.2.2 Eigenvalues in the Neighborhood of a Critical Point. Now, suppose that for some parameter value p_0, two uncoupled-blocked eigenvalues (for DOFs j and k, say) cross. The eigenvalues associated with the loci that cross are (to $O(\varepsilon^2)$) the solutions to

$$\begin{vmatrix} K_{jj}(p) - \lambda M_{jj} & -\varepsilon \\ -\varepsilon & K_{kk}(p) - \lambda M_{kk} \end{vmatrix} = 0 \tag{9}$$

while the remaining eigenvalues are still perturbations of $O(\varepsilon^2)$ from the uncoupled-blocked values. Taking a first-order Taylor series expansion of $K(p)$ about p_0 and putting

$$P = p - p_0; \quad \Lambda = \lambda - \lambda_0 \tag{10}$$

leads to

$$\begin{vmatrix} \alpha_j P - \Lambda M_{jj} & -\varepsilon \\ -\varepsilon & \alpha_k P - \Lambda M_{kk} \end{vmatrix} = 0 \tag{11}$$

where $\alpha = \partial K(p_0)/\partial p$. The eigenvalues of the coupled system are, therefore

$$\Lambda = \frac{1}{2}\left[(\beta_j + \beta_k)P \pm \sqrt{(\beta_j - \beta_k)^2 P^2 + \frac{4\varepsilon^2}{M_{jj}M_{kk}}}\right] \tag{12}$$

where

$$\beta_j = \frac{\alpha_j}{M_{jj}} = \frac{\partial \lambda_j^0}{\partial p} \tag{13}$$

The behavior is illustrated in Fig. 1: the critical point is the origin, where the uncoupled-blocked eigenvalue intersect. The solid lines show the eigenvalues of the weakly coupled system (Eq. (12)).

At $P = 0$, the eigenvalues of the coupled system are

$$\Lambda(0) = \pm\frac{\varepsilon}{\sqrt{M_{jj}M_{kk}}} \tag{14}$$

and thus the eigenvalue loci veer and do not cross. Their separation is $O(\varepsilon)$. Furthermore, the curvature of the eigenvalue loci are

$$\frac{\mathrm{d}^2\Lambda}{\mathrm{d}P^2} = \pm\frac{(\beta_j - \beta_k)^2}{\left((\beta_j - \beta_k)^2 P^2 + \frac{4\varepsilon^2}{M_{jj}M_{kk}}\right)^{\frac{3}{2}}}\frac{2\varepsilon^2}{M_{jj}M_{kk}} \tag{15}$$

This is a maximum at $P = 0$ where

$$\left(\frac{\mathrm{d}^2\Lambda}{\mathrm{d}P^2}\right)_{\max} = \pm\left(\frac{\sqrt{M_{jj}M_{kk}}(\beta_j - \beta_k)^2}{4}\right)\frac{1}{\varepsilon} \tag{16}$$

and the maximum curvature is thus $O(\varepsilon^{-1})$. Finally, $\Lambda = 0$ when the parameter P is

$$P = \pm\frac{1}{\sqrt{\alpha_j\alpha_k}}\varepsilon \tag{17}$$

2.2.3 Eigenvalues Far From a Critical Point. For values of p far from p_0 (so that $P^2 \gg 4m_2\varepsilon^2/m_1$), the eigenvalues are

$$\begin{aligned} \Lambda_j &\approx \beta_j P\left(1 + \frac{\varepsilon^2}{M_{jj}M_{kk}(\beta_j - \beta_k)\beta_j P^2}\right); \\ \Lambda_k &\approx \beta_k P\left(1 - \frac{\varepsilon^2}{M_{jj}M_{kk}(\beta_j - \beta_k)\beta_k P^2}\right) \end{aligned} \tag{18}$$

and are just perturbations of $O(\varepsilon^2)$ from the uncoupled-blocked values: note that the higher eigenvalue is slightly increased, the lower slightly decreased.

2.2.4 Eigenvectors. The eigenvectors are defined by the amplitudes of motion in the two coordinates (x_j, x_k). Far from the critical point the amplitude ratios can be written as

[1]Strictly, the terms ε should be written as $(\sqrt{K_{jj}K_{kk}}\varepsilon)$ for dimensional consistency: this is cumbersome, so the stiffness terms are omitted and the meaning of $O(\varepsilon)$ henceforth should be interpreted as defined in this footnote.

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$$\frac{\sqrt{M_{kk}}x_k}{\sqrt{M_{jj}}x_j} = -\frac{\varepsilon}{\sqrt{M_{jj}M_{kk}}(\beta_j - \beta_k)P};$$
$$\frac{\sqrt{M_{jj}}x_j}{\sqrt{M_{kk}}x_k} = \frac{\varepsilon}{\sqrt{M_{jj}M_{kk}}(\beta_j - \beta_k)P} \tag{19}$$

(the $\sqrt{M}$ terms are useful for mass-normalizing the uncoupled modes) and are thus small perturbations of $O(\varepsilon)$ from the uncoupled-blocked values 0. However, they change rapidly around the critical point, at which they become

$$\frac{\sqrt{M_{kk}}x_k}{\sqrt{M_{jj}}x_j} = -1; \qquad \frac{\sqrt{M_{jj}}x_j}{\sqrt{M_{kk}}x_k} = 1 \tag{20}$$

The eigenvectors rotate monotonically in the (x_j, x_k) plane as P increases with there being a "switching" effect around the critical point—see Fig. 1. Veering is thus associated with rapid changes (rotations) in the eigenvectors and a change in the eigenvectors associated with each branch of the eigenvalue spectrum.

2.2.5 Summary. In summary, if the uncoupled-blocked eigenvalues cross at $P = 0$, then the eigenvalues of the coupled system veer: they do *not* cross, equalling $\Lambda = \pm 2\varepsilon/\sqrt{M_{jj}M_{kk}}$ when $P = 0$. The veering occurs in a region of $O(\varepsilon)$ around the critical point (assuming $\alpha_{j,k} > O(\varepsilon)$) and the rate of change of the slopes of the eigenvalues is inversely proportional to ε: the weaker the coupling, the smaller is ε, the smaller the region in which veering occurs and the more rapid the veering. Veering is also associated with rapid rotation of the eigenvectors and the swapping of them between the two branches.

2.3 General Weak Conservative Coupling. The analysis above concerned weak stiffness coupling. The cases of weak mass or gyroscopic coupling, or any combination of the three, follow the same lines: the uncoupled-blocked system defines the skeleton, the intersections of which are the critical points; weak coupling results in a perturbation of the uncoupled-blocked eigenstructure and rapid veering around the critical points. For example, if there are stiffness, mass, and gyroscopic coupling terms $\varepsilon_k, \varepsilon_m, \varepsilon_g$, then the linearized system Eq. (11) become

$$\begin{vmatrix} \alpha_j P - \Lambda M_{jj} & -\varepsilon_k - \lambda_0\varepsilon_m - i\sqrt{\lambda_0}\varepsilon_g \\ -\varepsilon_k - \lambda_0\varepsilon_m + i\sqrt{\lambda_0}\varepsilon_g & \alpha_k P - \Lambda M_{kk} \end{vmatrix} = 0 \tag{21}$$

The above analysis holds with the term $(\varepsilon + \lambda_0\varepsilon_m)^2 + \lambda_0\varepsilon_g^2$ replacing ε^2 in the above equations. Equally, if there are variable mass parameters then the equations reduce to a similar form.

2.4 Strong Coupling. The case of strong coupling—when (some of) the off-diagonal coupling elements are the same order of magnitude as the diagonal elements—is not so straightforward. If veering is interpreted as a rapid change in the eigensolutions then veering does not occur in the presence of strong coupling: instead there is a gradual evolution of the eigenvalues rather than rapid veering around the critical points— ε is now $O(1)$. Furthermore, there is a distortion in the location of the points about which this evolution takes place from the critical points in the uncoupled-blocked skeleton. It could also be that the off-diagonal terms depend on the random parameters so that there is variability in both diagonal and off-diagonal terms. This inevitably involves strong coupling, since the variable parameter(s) (e.g., stiffness) and the off-diagonal terms are of the same order of magnitude as the diagonal terms. There is consequently again an evolution of the eigenvalue spectrum rather than rapid veering, and a distortion of the points in the (p, λ) plane about which the evolution takes place.

3 Degrees-of-Freedom, Models, and Continuous Systems

The analysis in Sec. 2 started, in Eq. (1), with a vector of amplitudes $\mathbf{x}$ of the DOFs. While in many cases there might be an "obvious" choice of DOFs, the analyst is of course free to choose any independent set of coordinates: there are many ways to select the DOFs and, while the system mass and stiffness matrices depend on the choice of DOFs, the eigenvalues of the system—and consequently the existence or not of veering—obviously do not depend on the analyst's selection. For example, a coordinate transformation of Eq. (1) can be made to a second set of coordinates $\mathbf{y}$, where $\mathbf{x} = \mathbf{P}\mathbf{y}$ and the matrices transform to $\mathbf{M}_y = \mathbf{P}^{\mathrm{T}}\ \mathbf{M}_x\ \mathbf{P}$ (and similarly for $\mathbf{K}_y$), where $\mathbf{M}_x$ is the mass matrix in terms of DOFs $\mathbf{x}$. The question then is whether *any* such coordinate transformation results in weak coupling in the sense of Eq. (3).

The issue is perhaps most important for continuous systems and for cases of practical interest. In practice, a discrete model of a continuous system is often made, for example, using finite elements or an assumed mode approach, and there are many potential choices of DOFs. For the case of continuous systems, various choices of DOFs might suggest themselves, particularly, when the system can be interpreted in terms of coupled subsystems, with the variable parameters $\mathbf{p}$ being localized within just one subsystem. Potential choices of the DOFs might be in terms of the modes of the individual subsystems. Two common examples are discussed in this section.

First, the DOFs $\mathbf{x}$ might be partitioned into sets associated with two subsystems a and b so that

$$\mathbf{x} = \begin{Bmatrix} \mathbf{x}_a \\ \mathbf{x}_b \end{Bmatrix} \tag{22}$$

where perhaps one subsystem involves a design variable, while the other does not (the parts of the skeleton associated with this subsystem are therefore constants). For example, in Sec. 5.3.3 the case of a beam, whose thickness is a variable parameter, resiliently coupled to a plate is considered: in this case $\mathbf{x}_a$ refers to the DOFs of the beam and $\mathbf{x}_b$ to those of the plate. The mass matrix of the structure can then be written as

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_a & 0 \\ 0 & \mathbf{M}_b \end{bmatrix} + \mathbf{M}_{c,x} \tag{23}$$

where the first term is block-diagonal, and the second represents coupling terms. The stiffness matrix can be written in a similar manner. Free vibration analyses of the eigenproblems $(\mathbf{K}_a - \lambda_a\mathbf{M}_a)\mathbf{x}_a = 0$ and $(\mathbf{K}_b - \lambda_b\mathbf{M}_b)\mathbf{x}_b = 0$ leads to mass-normalized modal matrices $\mathbf{P}_a, \mathbf{P}_b$, whose columns are the eigenvectors, while the uncoupled modal DOF vectors are $\mathbf{y}_a, \mathbf{y}_b$, where $\mathbf{x}_a = \mathbf{P}_a\mathbf{y}_a$, etc. Postmultiplication of the mass and stiffness matrices by

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_a & 0 \\ 0 & \mathbf{P}_b \end{bmatrix} \tag{24}$$

and premultiplication by its transpose yields

$$\mathbf{P}^{\mathrm{T}}\mathbf{M}\mathbf{P} = \begin{bmatrix} \mathbf{I} & 0 \\ 0 & \mathbf{I} \end{bmatrix} + \mathbf{P}^{\mathrm{T}}\mathbf{M}_{c,x}\mathbf{P};$$
$$\mathbf{P}^{\mathrm{T}}\mathbf{K}\mathbf{P} = \begin{bmatrix} \mathrm{diag}(\lambda_a) & 0 \\ 0 & \mathrm{diag}(\lambda_b) \end{bmatrix} + \mathbf{P}^{T}\mathbf{K}_{c,x}\mathbf{P} \tag{25}$$

This is in the form of Eq. (2). Note, however, that the coupling terms $\mathbf{M}_{c,y} = \mathbf{P}^{\mathrm{T}}\mathbf{M}_{c,x}\mathbf{P}$ and $\mathbf{K}_{c,y} = \mathbf{P}^{\mathrm{T}}\mathbf{K}_{c,x}\mathbf{P}$ depend on the physical system, the choice for the division into subsystems a and b and the choice of DOFs.

As a second possibility, the system might be divided into subsystems a and b which share common DOFs $\mathbf{x}_c$ on the interface between the subsystems—for example, a (stiff) beam rigidly connected to a (flexible) plate. Now, one likely coordinate transformation might be into the free interface modes of the beam (the

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stiff subsystem) and the fixed interface modes of the plate (the flexible subsystem). The DOFs can be partitioned into

$$\mathbf{x} = \begin{Bmatrix} \mathbf{x}_{a,i} \\ \mathbf{x}_c \\ \mathbf{x}_{b,i} \end{Bmatrix} \tag{26}$$

where the subscript i indicates internal DOFs. The subsystem matrices can be similarly partitioned, e.g.,

$$\mathbf{K}_a = \begin{bmatrix} \mathbf{K}_{a,ii} & \mathbf{K}_{a,ic} \\ \mathbf{K}_{a,ci} & \mathbf{K}_{a,cc} \end{bmatrix}; \quad \mathbf{K}_b = \begin{bmatrix} \mathbf{K}_{b,cc} & \mathbf{K}_{b,ci} \\ \mathbf{K}_{a,ic} & \mathbf{K}_{b,ii} \end{bmatrix} \tag{27}$$

The free interface modes of subsystem a are found from the eigenproblem $(\mathbf{K}_a - \lambda_a \mathbf{M}_a)\mathbf{x}_a = 0$, while the fixed interface modes of subsystem b follow from the eigenproblem $(\mathbf{K}_{b,ii} - \lambda_b \mathbf{M}_{b,ii})\mathbf{x}_b = 0$. The coupling stiffness matrix then becomes

$$\mathbf{K}_{c,y} = \mathbf{P}^T \begin{bmatrix} 0 & 0 & 0 \\ 0 & \mathbf{K}_{b,cc} & \mathbf{K}_{b,ci} \\ 0 & \mathbf{K}_{b,ic} & 0 \end{bmatrix} \mathbf{P} \tag{28}$$

with $\mathbf{P}$ again being given by Eq. (24) and with a similar expression for $\mathbf{M}_{c,y}$.

4 Nonconservative Systems

If there is damping, the eigenvalues become complex. As the variable parameter changes so the eigenvalues follow complex-valued loci. These may veer—characterized by the real parts of the eigenvalues moving apart and the eigenvectors rotating—but only if the damping is light enough. Otherwise the loci do not approach closely enough in the complex (P, Λ) space for veering to occur.

For a complex eigenvalue, free vibration is determined by the square roots of the eigenvalue. These are $\sqrt{\lambda} = \pm\omega_d + i\zeta\omega_n$ where $\omega_d = \omega_n\sqrt{1-\zeta^2}$ is the damped natural frequency while the imaginary part represents the effects of damping.

The critical point is now defined as that where undamped uncoupled-blocked eigenvalues intersect. Linearizing the equations about this point gives, for mass and gyroscopic coupling, for example,

$$\begin{vmatrix} \alpha_1 m_1 P + i m_1 \Delta_1 - \Lambda m_1 & -\lambda_0 \varepsilon_m - i\sqrt{\lambda_0}\varepsilon_g \\ -\lambda_0 \varepsilon_m + i\sqrt{\lambda_0}\varepsilon_g & \alpha_2 m_2 P + i m_2 \Delta_2 - \Lambda m_2 \end{vmatrix} = 0 \tag{29}$$

where $\Delta_{1,2}$ are the half-power bandwidths of the uncoupled-blocked systems. The solutions for the eigenvalues are

$$\Lambda = \frac{1}{2}\Big((\alpha_1+\alpha_2)P + i(\Delta_1+\Delta_2) \pm \sqrt{\big((\alpha_1-\alpha_2)P + i(\Delta_1-\Delta_2)\big)^2 + 4\frac{\varepsilon^2}{m_1 m_2}}\Big) \tag{30}$$

where $\varepsilon^2 = \lambda_0^2\varepsilon_m^2 + \lambda_0\varepsilon_g^2$, and thus, the eigenvalues are complex. At $P = 0$ the eigenvalues are

$$\Lambda = \frac{1}{2}\left(i(\Delta_1+\Delta_2) \pm \sqrt{4\frac{\varepsilon^2}{m_1 m_2} - (\Delta_1-\Delta_2)^2}\right) \tag{31}$$

Veering in the eigenvalues thus occurs if

$$4\frac{\varepsilon^2}{m_1 m_2} > (\Delta_1 - \Delta_2)^2 \tag{32}$$

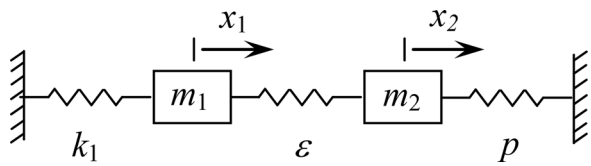

Fig. 2 2DOF weakly coupled system

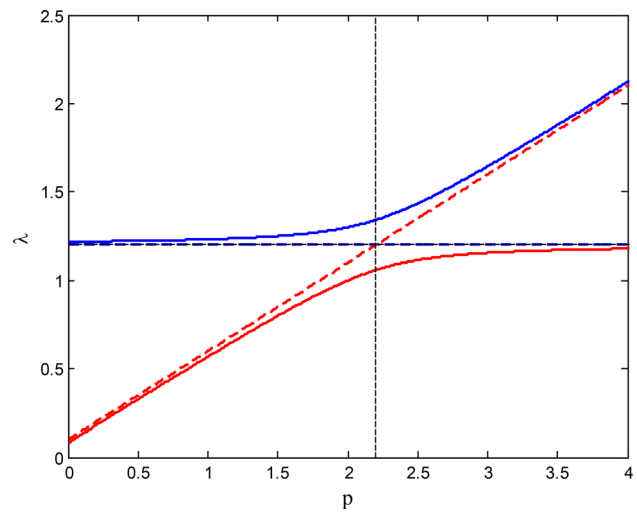

Fig. 3 Eigenvalue loci for the 2DOF system of Fig. 2: and skeleton of the uncoupled-blocked system; and eigenvalues of the coupled system; - - - position of the critical point

that is, if the coupling is large enough compared to the *difference* of the bandwidths of the uncoupled blocked system. A numerical example is shown in Sec. 5.1.2.

5 Numerical Examples

5.1 Two-Coupled Oscillators

5.1.1 Undamped Coupled Oscillators—Weak Coupling. Consider first the illustrative example of two, undamped, coupled oscillators shown in Fig. 2. The stiffness ε of the spring coupling the two masses is assumed to be small, while the spring of stiffness p is a variable parameter. This might also represent the coupling between two modes, the DOFs being modal coordinates. The uncoupled-blocked eigenvalues are $(k_1+\varepsilon)/m_1$ and $(p+\varepsilon)/m_2$. The critical point, where these two eigenvalues are equal, is where $p = (m_2 k_1 + (m_2 - m_1)\varepsilon)/m_1$. Figure 3 shows the skeleton (the eigenvalues of the uncoupled system) and the eigenvalue loci of the coupled system, using the values $m_1 = 1; m_2 = 2; k_1 = 1; \varepsilon = 0.2$. Veering occurs around the critical point at $p = 2.2$. Figure 4 shows the slope $d\lambda/dp$ and the curvature (sensitivity) $d^2\lambda/dp^2$ of the eigenvalues. It can be seen that the curvature reaches a maximum at the critical point. Veering can also be seen by noting the rotation of the eigenvectors (or the rate of rotation of the eigenvectors) when the parameter varies. The directions of the eigenvectors, defined by the angles $\theta = \tan^{-1}(\sqrt{m_2}x_2/\sqrt{m_1}x_1)$, for the coupled system are shown in Fig. 5: they are of course perpendicular, and the directions associated with each eigensolution can be seen to rotate rapidly around the critical point, being ± 45 deg at that point, and with the directions associated with each branch far below and far above the critical point being exchanged.

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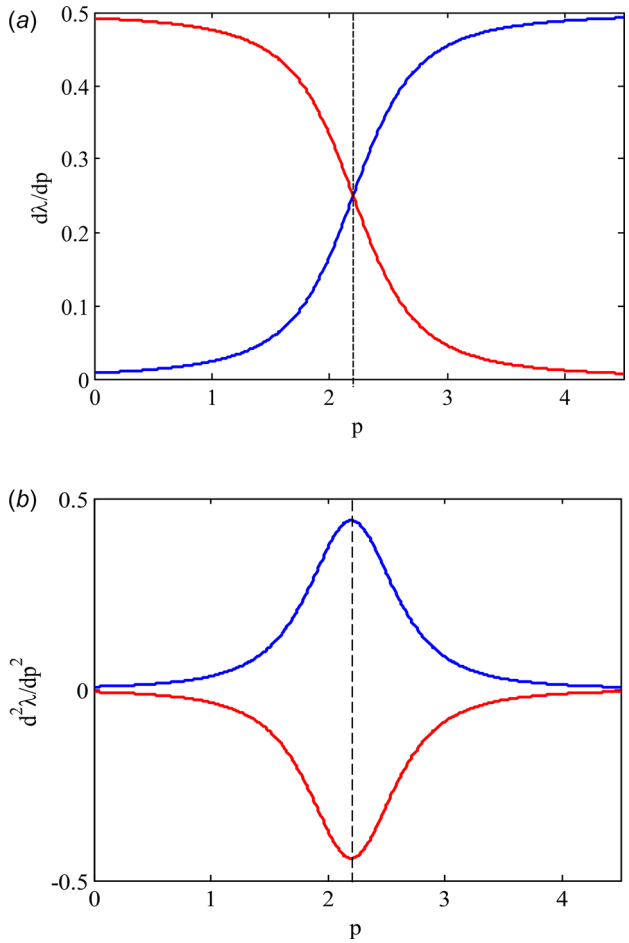

Fig. 4 (*a*) Slope and (*b*) curvature of the eigenvalue loci in the 2DOF system of Fig. 2; – – – – – position of the critical point

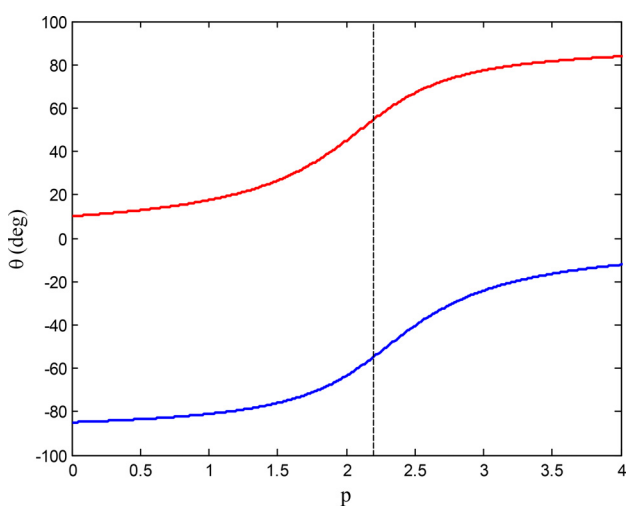

Fig. 5 Eigenvector angle in the 2DOF system of Fig. 2; - - - position of the critical point

5.1.2 Undamped Coupled Oscillators—Strong Coupling. Consider again the 2DOF system in Fig. 2, but assume now that the variable parameter is the stiffness ε of the coupling spring, while the stiffness $p = k_2$ is given and constant. In this case, both the diagonal and off-diagonal terms in Eq. (2) depend on the variable parameter and strong coupling occurs. Figure 6 shows the eigenvalue loci when $m_1 = 1; m_2 = 2; k_1 = 1; p = k_2 = 3$. The

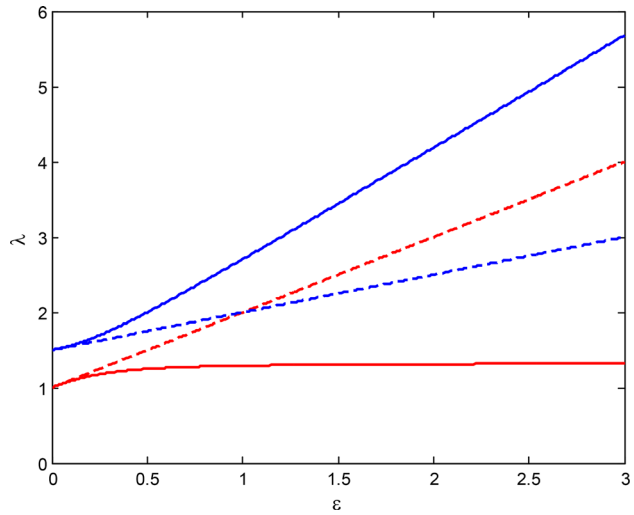

Fig. 6 Strong coupling in the 2DOF system of Fig. 2. Note a gradual evolution of the eigenvalues of the coupled system as opposed to veering around the critical points.

critical point is now at $\varepsilon = 1$. Because the coupling is strong, there is now a rather gradual evolution of the eigenvalue loci rather than rapid veering about the critical point.

5.1.3 Damped Coupled Oscillators. The system of Sec. 5.1.1 and Fig. 2 is considered, but damping is now included. The masses are coupled through a weak spring ε, and the variable parameter is the stiffness p. The same parameters as in Sec. 5.1.1 are used, and hence weak coupling is considered. The half-power bandwidths of the uncoupled-blocked system are $\Delta_1 = 0.25$ and $\Delta_2 = 0.1$ so that the condition in Eq. (32) holds. Figure 7 shows the real and imaginary parts of the frequency loci in the (p, ω) plane. It can be seen that veering occurs as described in Sec. 4. The eigenvalue loci in the (P, Λ) plane show a similar behavior to those depicted in Figs. 3 and 4, and the imaginary parts of the eigenvalue are equal at the critical point $P = 0$, see Eq. (31).

5.2 Multiple Degrees-of-Freedom System

5.2.1 Weak Coupling. In this section, the case of a MDOFs spring-mass chain is considered. There are n masses, the ith mass m_i is connected to its neighbors by springs of stiffness k_i and k_{i+1} and to ground by an additional spring k_0 as shown in Fig. 8. In the numerical example, the spring k_n is small (i.e., weak coupling between masses m_{n-1} and m_n) and $k_{n+1} = p$ is the variable parameter. This system thus consists of an $(n-1)$ DOF subsystem with fixed parameters weakly coupled (through the spring k_n) to an oscillator m_n with a variable stiffness p.

The uncoupled-blocked eigenvalues for the subsystem with fixed parameters are found by an eigenanalysis of the system comprising masses 1 to $n-1$ and springs 1 to n—these do not depend on p. For mass m_n the uncoupled-blocked eigenvalue is

$$\lambda_n^{(0)} = \frac{k_n + k_{n+1}}{m_n} = \frac{k_n + p}{m_n} \tag{33}$$

These form the skeleton shown in Fig. 9(*a*) for a 6DOF system with $m_i = 1$; $k_0 = 1$; $k_i = k_0(1+i/6)/2$; $k_n = k_0/4$. The critical points are the intersections of these uncoupled-blocked eigenvalues. The eigenvalues of the coupled system, Fig. 9(*b*), veer around the critical points. The eigenvectors of the lowest mode for small p (<0.5) and the highest mode for large p (>4.5) are essentially identical, involving motion predominantly of mass m_n.

5.2.2 Strong Coupling. For the 6DOF system considered in Sec. 5.2.1, strong coupling occurs when $k_n = 2k_0$ is no longer

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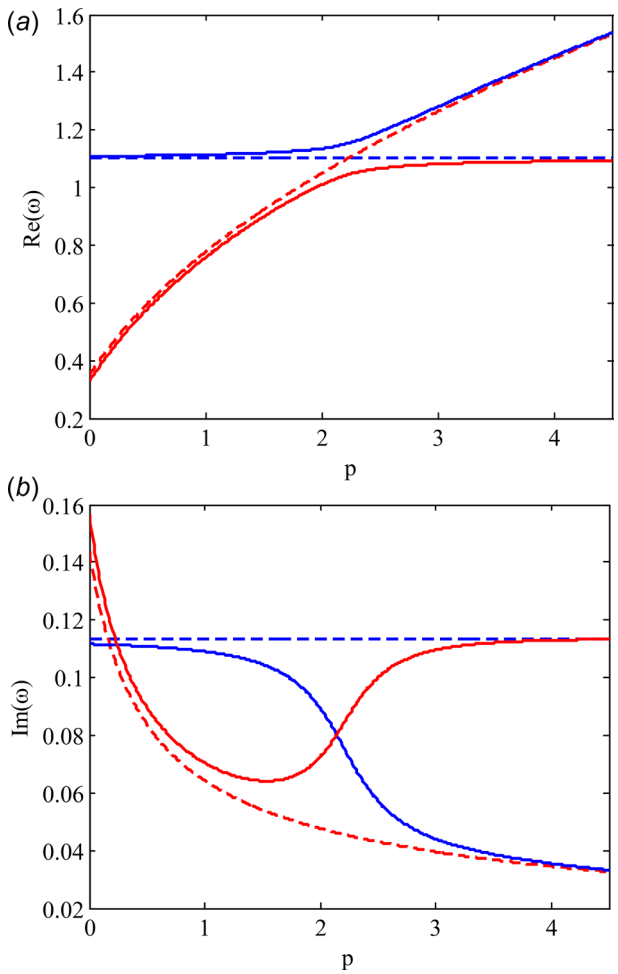

Fig. 7 (*a*) Real and (*b*) imaginary parts of the natural frequency loci for the 2DOF system of Fig. 2 in the (P, ω) plane: and skeleton of the uncoupled-blocked system; and frequencies of the coupled system

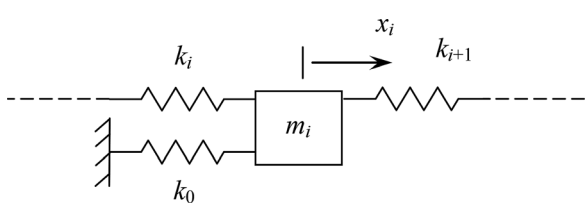

Fig. 8 The *i*th mass in a mass-spring chain of *n* masses

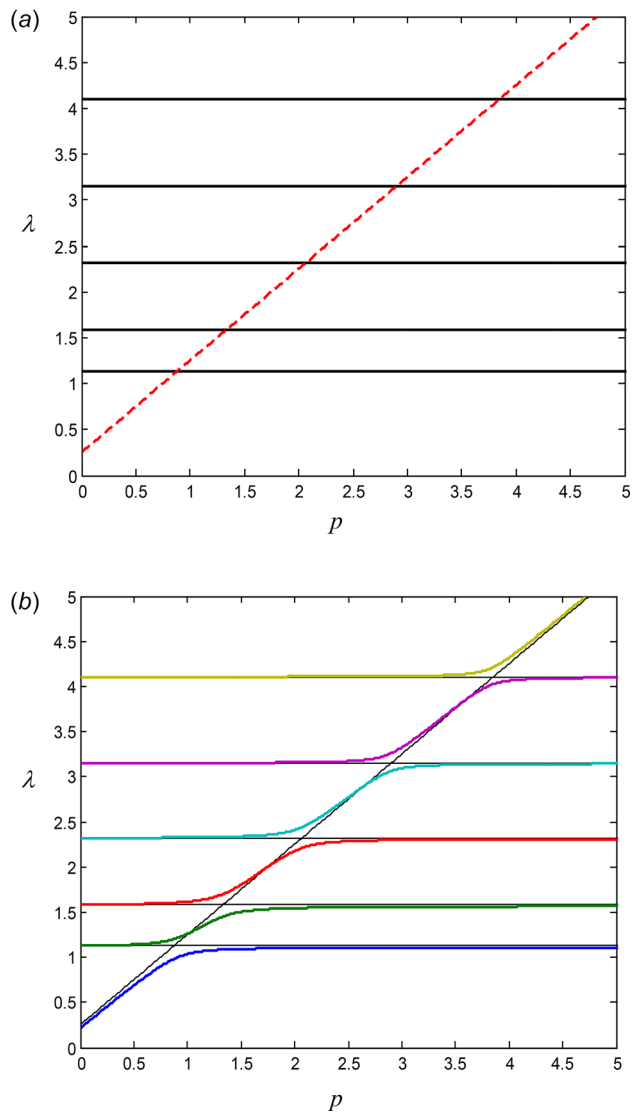

Fig. 9 Veering in a 6DOF system: $m_i = 1$; $k_0 = 1$; $k_i = k_0(1+i/6)/2$; $k_n = k_0/4$; k_{n+1} is a variable stiffness. (*a*) Skeleton: eigenvalues of the uncoupled-blocked system, eigenvalues of fixed system; eigenvalue of system comprising m_n, k_n, and k_{n+1}. Note intersections at the critical values of *p*. (*b*) Eigenvalues of coupled system and skeleton. Note rapid veering around the critical points.

small. In this case, the eigenvalue loci now show gradual evolution rather than rapid veering around the critical points.

5.3 Continuous Systems

5.3.1 Plate Coupled to a Variable Oscillator—Weak Coupling. The system comprises a rectangular simply supported plate to which a 1DOF oscillator is attached at point (x_0, y_0) through a weak coupling spring of stiffness ε as shown in Fig. 10. The stiffness p of the oscillator is variable. The amplitudes y_j of the modes of vibration of the plate alone (i.e., when $\varepsilon = 0$) are taken as the DOFs of the plate. The *j*th mass-normalized mode shape is

$$\phi_j(x, y) = \frac{2}{\sqrt{\rho h L_x L_y}} \sin \frac{m_j \pi x}{L_x} \sin \frac{n_j \pi x}{L_y} \tag{34}$$

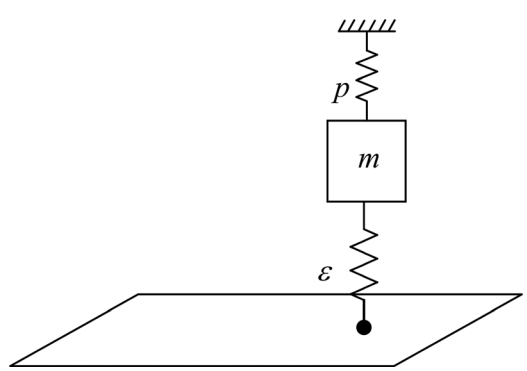

Fig. 10 Rectangular simply supported plate with an attached oscillator

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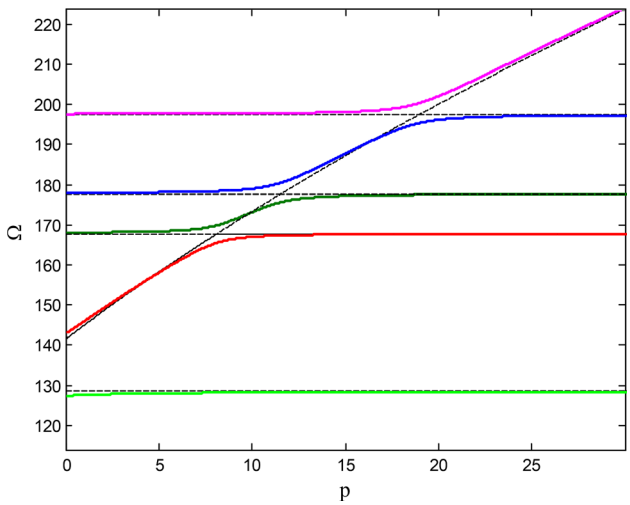

Fig. 11 Veering in a rectangular simply supported plate with an attached oscillator: $L_y/L_x = 1.5$; $m_a = 0.008$; $\varepsilon_a = 20$; $(\bar{x}, \bar{y}) = (0.43, 0.22)$; ▬▬▬ eigenvalues of the coupled system; - - - - - eigenvalues of the uncoupled-blocked system

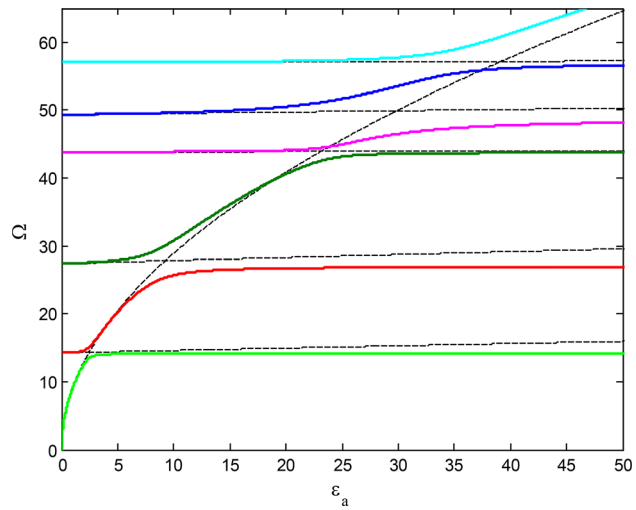

Fig. 12 Strong coupling: veering in the plate system of Sec. 5.3.2; ▬▬▬ eigenvalues of the coupled system; - - - - - eigenvalues of the uncoupled-blocked system. Note that a gradual evolution of the eigenvalues of the coupled system as opposed to veering around the critical points occurs for increasing values of ε.

where m and n are integers, ρ is the material density, and L_x and L_y are the lengths of the sides of the plate. The uncoupled-blocked eigenvalues of the plate are to a very good approximation

$$\lambda_{p,j}^{(0)} = \lambda_{p,j} + \varepsilon\phi_j^2(x_0, y_0) \tag{35}$$

where $\lambda_{p,j}$ is the jth eigenvalue of the plate alone. The uncoupled-blocked eigenvalue of the oscillator is

$$\lambda_o^{(0)} = \frac{p+\varepsilon}{m} \tag{36}$$

When coupled, the mass matrix is diagonal but there are small off-diagonal terms $-\varepsilon\phi_i(x_0, y_0)\phi_j(x_0, y_0)$ in the stiffness matrix which couple blocked modes i and j in the plate.

Figure 11 shows the loci of the natural frequencies of the uncoupled-blocked and coupled systems as functions of p when $L_y/L_x = 1.5; m = 0.008; \varepsilon = 20; (\bar{x}, \bar{y}) = (0.43, 0.22)$, where $\bar{x} = x_0/L_x$ and $\bar{y} = y_0/L_y$. The nondimensional parameters

$$\Omega = \omega L_x^2\sqrt{\frac{\rho h}{D}}; \quad \varepsilon_a = \frac{\varepsilon {L_x}^2}{D}; \quad m_a = \frac{m}{\rho h L_x L_y}$$

are introduced. Again it can be noticed that far from the critical points the natural frequencies are almost those of the uncoupled system, while close to the critical points the eigenvalues veer.

5.3.2 Plate Coupled to a Variable Oscillator—Strong Coupling. Figure 12 shows the same example as in Sec. 5.3.1, except that the coupling spring is now the variable parameter and the spring connecting the mass to ground is removed. It can be noticed that for small values of ε weak coupling occurs, while as ε increases changes in the eigenvalues become less rapid. The case of larger ε involves strong coupling, since the diagonal and the off-diagonal terms are of the same order of magnitude as the diagonal terms.

5.3.3 Beam-Plate System—Weak Coupling and Strong Coupling. In this section, the bending vibrations of a system consisting of a simply supported Euler–Bernoulli beam and a simply supported rectangular thin plate are considered. The plate and beam are connected (coupled) together by a resilient layer of stiffness per unit length ε along the line $y = y_0$ as shown in Fig. 13.

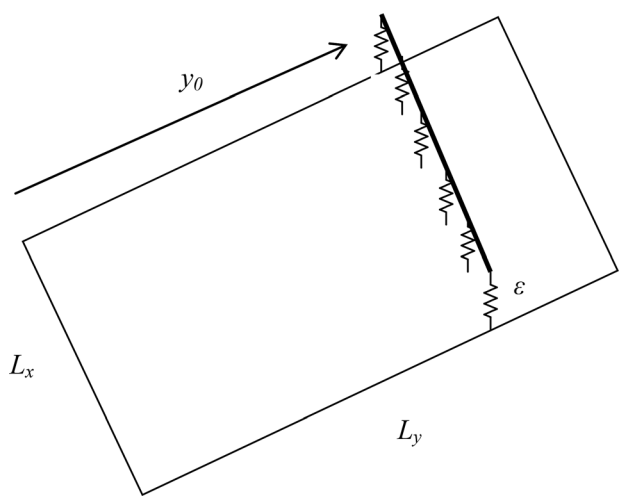

Fig. 13 Simply supported plate coupled to a simply supported beam by a resilient layer of stiffness per unit length ε

The variable parameter considered in this example is the thickness h_b of the beam.

A discretized model is developed using the assumed modes method, taking as the DOFs the modes of the beam and the plate alone. An alternative approach would be to use a standard finite element analysis. The mode shapes $\phi_j(x, y)$ of the plate and $\varphi_i(x)$ of the beam are given by

$$\phi_j(x, y) = \sin\frac{m_j\pi x}{L_x}\sin\frac{n_j\pi x}{L_y}; \quad \varphi_i(x) = \sin\frac{m_i\pi x}{L_x} \tag{37}$$

where m and n are integers, and L_x and L_y are the lengths of the sides of the plate. The uncoupled-blocked eigenvalues of the plate and the beam are to a very good approximation

$$\begin{aligned} \lambda_{p,j}^{(0)} &= \lambda_{p,j} + \varepsilon\int_0^{L_x}\phi_j^2(x, y_0)dx \\ \lambda_{b,i}^{(0)} &= \lambda_{b,i} + \varepsilon\int_0^{L_x}\varphi_i^2(x)dx \end{aligned} \tag{38}$$

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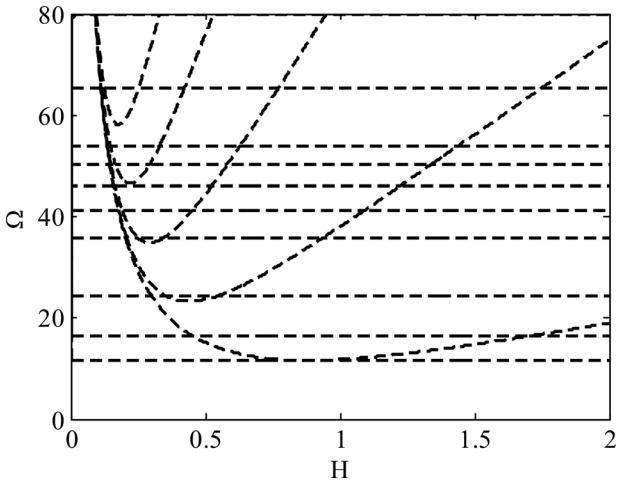

Fig. 14 Simply supported plate coupled to a simply supported beam by a resilient layer of stiffness ε: $L_y/L_x = 42/17$; $y_0 = L_y/6$; - - - - - eigenvalues of the uncoupled-blocked system. Note intersections at the critical values.

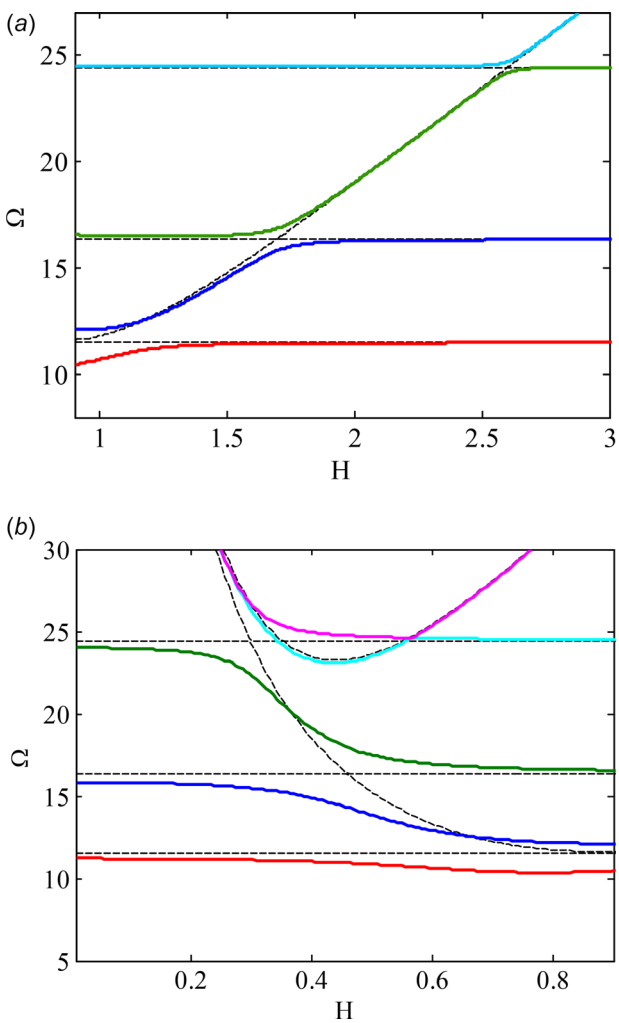

Fig. 15 Simply supported plate coupled to a simply supported beam by a resilient layer of stiffness ε: (*a*) weak coupling and (*b*) strong coupling; $L_y/L_x = 42/17$; $y_0 = L_y/6$. ——— eigenvalues of the coupled system; - - - - - eigenvalues of the uncoupled-blocked system

where $\lambda_{p,j}$ and $\lambda_{b,i}$ are the jth and the ith eigenvalues of the uncoupled plate and the beam, respectively.

When the beam and the plate are coupled, there are off-diagonal terms $-\varepsilon \int_0^{L_x} \varphi_i(x)\phi_j(x, y_0)dx$ in the stiffness matrix. Depending on the thickness of the beam, which is taken to be the variable parameter, these off diagonal terms can be either small or comparable (in magnitude) to the diagonal terms. Thus, weak or strong coupling can occur. The nondimensional parameters $\Omega = \omega L_x{}^2\sqrt{(\rho h_p)/d}$; $H = (h_b/h_p)$ are introduced, where ρ is the material density, D is the stiffness of the plate, and h_b and h_p are the thicknesses of the beam and of the plate, respectively.

In this example the values $L_y/L_x = 42/17; y_0 = L_y/6$ are assumed. The skeleton of the uncoupled blocked system is shown in Fig. 14. For this example, the uncoupled blocked plate natural frequencies do not of course depend on the thickness of the beam. The uncoupled blocked natural frequencies of the beam depend on the thickness, which affects both the mass and stiffness. For low values of thickness, the natural frequencies decrease as $h_b{}^{-1/2}$, the stiffness of the beam being negligible to that of the line support. The natural frequencies become proportional to h_b as the thickness increases, since the stiffness of the line support becomes negligible. Typically, the coupling becomes weak for large enough beam thickness.

Figures 15(*a*) and 15(*b*) show the eigenvalue loci in two ranges of the beam thickness, where the coupling is weak (larger beam thickness, when there is rapid veering around the critical points and the eigenvalue loci are close to the skeleton of the uncoupled-blocked system) and strong (smaller beam thickness, when there is a more gradual evolution).

6 Concluding Remarks

This paper concerned mode veering in systems with varying parameters. A general discrete model was considered, although the analysis applies equally to discrete MDOF models of continuous systems. First, weak coupling was defined such that the off-diagonal coupling terms in the system matrices are small compared to the diagonal terms in the sense of Eq. (3). The uncoupled-blocked system was then defined to be that where all DOFs but one are blocked. The uncoupled-blocked eigenvalues as functions of the variable parameters then form the skeleton of the system. The loci of these eigenvalues intersect at the critical points. For weakly coupled systems veering was seen to comprise rapid changes of the eigenvalues with respect to the variable parameters, in small regions of the parameter-eigenvalue space around the critical points: for coupling terms of order ε veering occurs in a region of order ε around the critical points, with the rate of change of eigenvalues being of order ε^{-1}. This is accompanied by rapid rotations in the eigenvectors in this small region where veering occurs, with them swapping over between the two branches of the skeleton on either side of the critical point. Far from the critical point the coupling introduces perturbations of the eigenvalues of order ε^2 (the higher eigenvalue increases, the lower decreases), while the eigenvectors are perturbed by order ε.

While initially attention was focused on stiffness coupling, veering was seen to occur for stiffness, mass and/or gyroscopic coupling in conservative systems. For damped systems, it was seen that veering occurs if the difference between the levels of damping in the two (uncoupled-blocked) DOFs is small enough. When modeling a structure, various choices of DOFs could be made, and veering occurs if any choice results in a weakly coupled system in the sense defined. This is most pertinent when developing discrete models of continuous systems, whether through finite element analysis, assumed mode models, component modal analysis or whatever. Numerical examples were presented.

The phenomenon of veering occurs widely in bifurcation behavior of eigenvalue problems involving arbitrarily weak coupling, including modal analysis, wave propagation, and stability analysis, and the conclusions drawn here apply equally to those

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other situations.[2] The purpose of this paper was partly to describe the general circumstances concerning whether or not veering occurs. The definitions of uncoupled-blocked system, the skeleton, and the critical points identify where in parameter space veering might be important, and these might be able to be calculated at little cost. At these points, the behavior of the system becomes particularly sensitive to the system parameters, depending on the coupling terms. Apart from this general interest, veering can have consequences for practical applications. Design often involves placing constraints on the natural frequencies of the system or on its components. Often this might be to ensure adequate spacing (i.e., mismatch) although sometimes the aim might be for two components to have coincident natural frequencies: in vibration energy harvesting, for instance, the energy harvesting mechanism might be resonant and tuned to a particularly natural frequency of the host structure. Coupling is required for energy transfer, so the coupled natural frequencies necessarily veer and depend sensitively on the system parameters. In model updating or parameter estimation from modal data, the existence of veering will have consequences for the sensitivity of properties on modal data. In the analysis of systems with uncertain or variable properties, frequency or ensemble average response prediction are often made, together with an estimate of variance. These are known to depend on the natural frequency statistics of the system or its component subystems. Veering affects this distribution: the "mode repulsion" that occurs between two closely spaced natural frequencies is a consequence of the increasing sensitivity as the natural frequencies approach and the coupling between them gets weaker.

Appendix: Evaluation of the Determinant

Consider the matrix $\mathbf{D} = \mathbf{K} - \lambda\mathbf{M}$. Assuming that the coupling is weak in the sense defined, then, the off-diagonal terms in $\mathbf{K}$ and $\mathbf{M}$ are of order ε compared to the diagonal terms d_{jj}. The determinant D can be found by expanding in terms of cofactors, following any row or column. For example, expansion following the first row gives [34]

$$D = \sum_{j=1}^{n} d_{1j}\mathrm{cof}(d_{1j}) \tag{A1}$$

where $\mathrm{cof}(d_{1j})$ is the cofactor of d_{1j}. Now $\mathrm{cof}(d_{11})$ contains a term involving the product of the diagonal terms, the other terms being $O(\varepsilon^2)$ or smaller. For an off-diagonal term, the largest term in $\mathrm{cof}(d_{1j})$ is $O(\varepsilon)$ (i.e., the term involving the product of the diagonal terms except for those in row 1 and row j). Since d_{1j} is $O(\varepsilon)$, then, the contribution of the j'th term in Eq. (A1) to the determinant is $O(\varepsilon^2)$ or smaller, leading to Eq. (4).

Close to a critical point two of the diagonal elements, say d_{11} and d_{22}, are also small. The cofactors or row 1 become

$$\begin{aligned}
\mathrm{cof}(d_{11}) &= d_{22}\left(\prod_{i=3,n} d_{ii} + O(\varepsilon)\right) + O(\varepsilon^2)\\
\mathrm{cof}(d_{12}) &= -d_{21}\left(\prod_{i=3,n} d_{ii} + O(\varepsilon)\right) + O(\varepsilon^2)\\
\mathrm{cof}(d_{1i}) &= O(\varepsilon^2)
\end{aligned} \tag{A2}$$

The determinant then becomes

$$D = (d_{11}d_{22} - d_{12}d_{21})\left(\prod_{i=3,n} d_{ii} + O(\varepsilon)\right) + O(\varepsilon^2) \tag{A3}$$

[2]A proviso: here, as in the veering of dispersion curves in Ref. [33], the rates of change of the two intersecting branches of the skeleton have the same slope. If they have opposite slope, locking [33] or dynamic instability (flutter) occurs.

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